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Commands for navigating the file system

LS

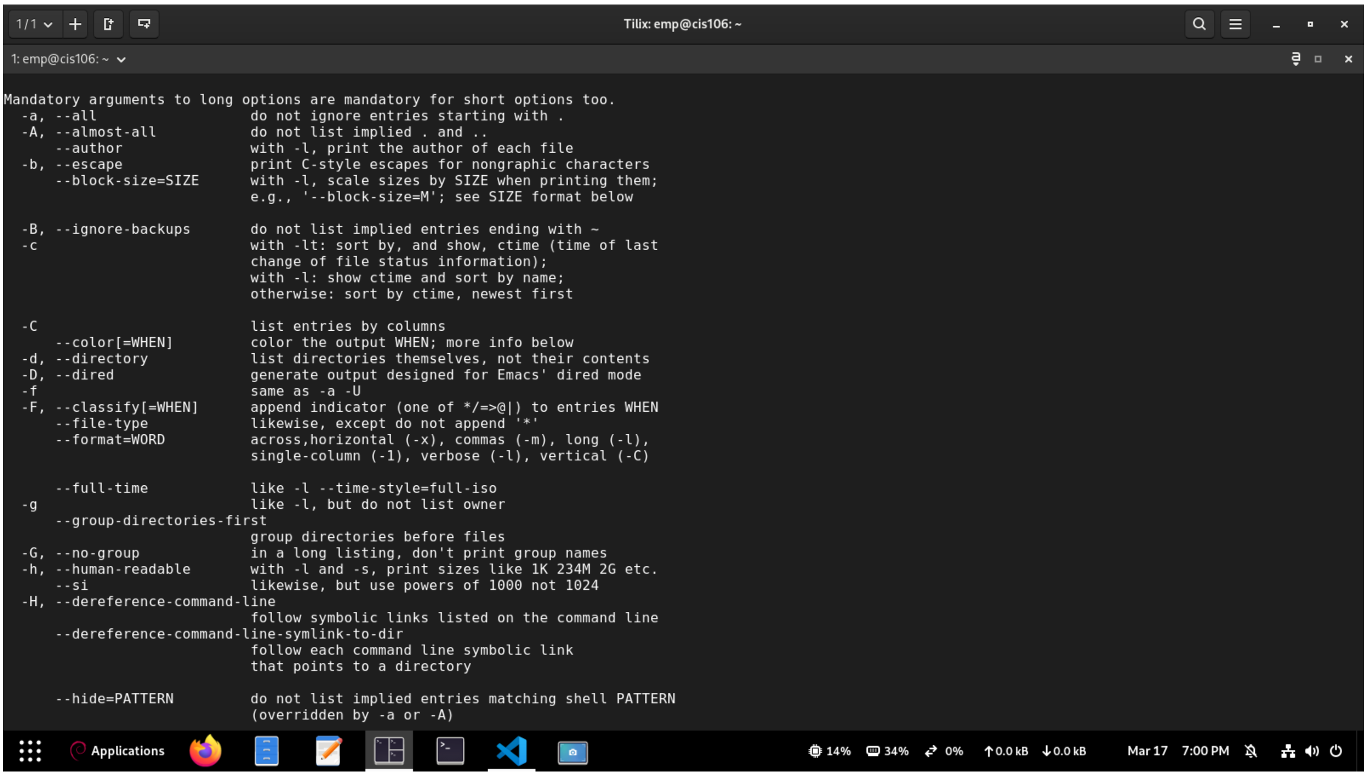
- ls is a command used to display files and folders in a directory.
- If no directory is given, it will show the contents of the current location by default.
- Listing means checking what items exist inside a directory.
- I use this command whenever I want to quickly view what's inside a folder on my Linux system.
- It can also be used to view file details (metadata), such as last modified time, file size, and permissions.

FORMULA/SYNOPSIS

- `ls` + `option` + `directory(ies) or file to list`

Examples

- List all files, including hidden ones, in long format:
- `ls -la`
- List files in a specific directory with human readable sizes
- `ls -lh ~/Documents`
- Recursively list all files in current directory and subdirectories:
- `ls -R`



CD

The `CD` command, short for change directory, is a fundamental Linux command used to navigate the file system within the terminal

Description

-`cd` + `path do directory`

Examples

- Change to a specific directory using an absolute path:
- `cd /home/user/Documents`
- Move up one directory level (parent directory):
- `cd ..`
- Return to your home directory:
- `cd`

PWD

The pwd command in Linux stands for`"print working directory".`When executed in the terminal, it displays the full, absolute path of your current location within the filesystem hierarchy, starting from the root directory.

FOMULA/SYNOPSIS

- `pwd`

Examples

- Display the full path of the current directory:
- `pwd`
- Use with `cd` to see your location before moving to another directory:
- `pwd`
- Combine with echo to store the current path in a variable:
- `current_dir=$(pwd)`

What is a variable?

A variable is a named storage location in a program or shell that holds a value. It can store text, numbers, or other types of data and can be used to refer to that data later.

How do I use a variable?

You use a variable by assigning a value to it and then referencing it when needed. For example, in a shell: `name="Eric" echo $name` This will output Eric.

What is an environment variable?

An environment variable is a variable that is available system-wide or to all processes started from a shell. It is often used to store settings that affect the behavior of software, like PATH or HOME.

What is a user defined variable?

A user defined variable is a variable that you create yourself in a shell or program. Unlike environment variables, it only exists in your current session unless exported.

What is the root directory?

The root directory is the top-most directory in a file system, represented by / in Linux. All other directories and files branch from the root directory.

What does “Parent Directory” mean?

A parent directory is the directory that contains the current directory. For example, if you are in /home/user/Documents, the parent directory is /home/user.

What is an absolute path? Include an example

An absolute path is the full path from the root directory to a file or directory. It always starts with /.

- Example: `/home/user/Documents/file.txt`

What is a relative path? Include an example

A relative path is a path that is relative to your current working directory. It does not start with /.

- Example: `If you are in /home/user, the relative path to the Documents folder is Documents/file.txt`

What is the difference between “Your home directory” and “The home directory”?

“Your home directory” is the personal directory assigned to you, usually /home/username. “The home directory” can refer to any user's home directory or the general concept of home directories on the system.